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you name them one by one, it will take too much time. But if you can select them all and use the auto-number function, resistors get incremented reference numbers automatically. Such functions save time and reduce human error, so do look for them in any EDA tool you are reviewing.

Supported image formats. Often, you will have to publish your schematic, for which you need it in some image format such as EPS, JPEG, etc. Do check if the image format required by you is supported by the EDA.

Configurability. All EDAs are designed corresponding to standard needs but every organisation has unique requirements for an EDA. For example, one organisation might need a different colour-map for exported images or a BOM in a different format, and so on. Here, configuration comes into play. In proprietary tools, configuration access is only on the surface level and you can change only a very limited number of things. But several open source tools are completely configurable and you can change them to suit your needs. So if your organisation has such unique requirements which no proprietary software supports, then open source tools are what might help.

Exporting designs to the PCB tool

The EDA tool you select should support the complete design cycle. Once the schematic entry is done, it is simulated for verification. But as per past experiences, there is always some gap between the simulated and real result. Therefore prototyping is done and all the tests are performed on the physical sample. All electronic assemblies are done on the PCB board, so you have to

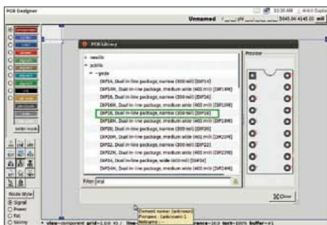


Fig.4: A PCB design tool

make the PCB layout for your design. EDA tools should have the function that enables your design to be exported to the PCB design tool in an intuitive manner.

Some PCB tools even accept the netlist generated by other schematic capture tools for PCB layout. If you plan to work that way, check what netlist formats are accepted by the tool. Also, your organisation might outsource the PCB design work. So it is important that your schematic capture tool can generate a netlist compatible to other tools as well.

The PCB design tool

The PCB design tool is used to make the PCB layout of your design. The PCB is manufactured with this layout and components are mounted on it to build the prototype. PCB design is creative work, so the tool should be very efficient in handling repetitive cumbersome jobs and should be extremely user friendly. Also, this is one task in the design phase that requires a lot of concentration and has maximum chances of human error. So the tool should be able to automatically indicate the human errors.

Similar to schematic capture tool, one of the key parameters to consider while selecting the PCB tool are the available in-built footprint library and quick footprint creation. The more complete the footprint library is, the less time you will spend making new footprints. Unlike symbols, footprints